

# 1 Topological Spaces

## Definition 1.1: Topology

Let  $X$  be any set. A topology  $T$  on  $X$  is a collection of subsets of  $X$ , which satisfies the following axioms.

- $\emptyset$  and  $X$  are both in  $T$
- Closed under arbitrary unions
- Closed under finite intersections

The sets in  $T$  are called open sets.

A set  $X$  equipped w/ a topology is called a topological space.

*Example:*

text

- $X$  is any set,  $T = P(X)$ . (discrete topology)
- $X$  is any set,  $T = \{\emptyset, X\}$ . (indiscrete topology)
- $X = \{a, b, c\}$ ,  $T = \{\emptyset, X, \{a, b\}\}$
- $X = \mathbb{R}$ ,  $T = \{\emptyset, X\} \cup \{U \subset \mathbb{R} \mid \text{for all } x \in U, \exists \epsilon > 0 \text{ s.t. } (x - \epsilon, x + \epsilon) \subset U\}$
- $X$  is any set, let  $T$  be the collection of subsets  $U \subseteq X$  s.t. either  $U = \emptyset$ , or  $X \setminus U$  is finite.

## Proposition 1.2

This is a topology.

*Proof.*

- $\emptyset, \mathbb{R}$  are both open
- Suppose  $\{U_j\}_{j \in J}$  is a collection of open sets in  $\mathbb{R}$   
Let  $x \in \bigcup_{j \in J} U_j$ . Then  $x \in U_j$  for some  $j \in J$ .  
 $U_j$  is open thus there is an open interval in  $U_j$  containing  $x$ . So it is in the arbitrary union
- Let  $U, V$  be open subsets of  $\mathbb{R}$

Let  $x \in U \cap V$ . Because  $U, V$  open there are intervals  $(a, b) \subset U$ ,  $(c, d) \subset V$  s.t.  $x \in (a, b) \cap (c, d)$   
 $(a, b) \cap (c, d) = (\max(a, c), \min(b, d)) \subset U \cap V$ . Thus,  $x$  is in this interval thus  $U \cap V$  is open.

### Proposition 1.3

The "cofinite" topology is a topology.

*Proof.*

- The empty set is open, and  $X \setminus X = \emptyset$  is finite, so  $X \in T$
- Consider sets  $\{U_j\}_{j \in J}$  s.t.  $U_j \in T$   
 First: consider the case for some  $j'$ ,  $X \setminus U_{j'}$  is finite.  
 $X \setminus \bigcup_{j \in J} U_j = \bigcap_{j \in J} X \setminus U_j \subset X \setminus U_{j'}$ . So the arbitrary union is finite.  
 Second: If the first case does not occur, then  $U_j = \emptyset$  for all  $j \in J$ , so the union is the empty set.
- Let  $U, V \in T$ . If either is the empty set then  $U \cap V = \emptyset \in T$ , so assume that both are not empty. Then consider  $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ , so it is a union of two finite sets which is finite thus  $U \cap V \in T$

### Definition 1.4: Closed Set

If  $X$  is a set with topology  $T$ , then the complement of an open set in  $X$  is called a closed set.

*Example*

If  $T$  is cofinite topology, a set in  $A \subseteq X$  is closed  $\iff$  either finite or  $A = X$

**IMPORTANT:** Some sets are neither open nor closed. Some sets are both closed and open.

## 2 Basis for a topology

### Definition 2.1: Basis

Let  $X$  be a set. A collection  $\beta$  of subsets of  $X$  is a basis for a topology if:

- every  $x \in X$  is in some  $B \in \beta$
- If  $x \in B_1 \cap B_2$  for  $B_1, B_2 \in \beta$   
then:  $\exists B_3 \in \beta$  s.t.  $x \in B_3 \subseteq B_1 \cap B_2$

The topology generated by  $\beta$  is the collection  $T$  of subsets defined by:

$$U \in T \iff \text{for every } x \in U, \exists B \in \beta \text{ s.t. } x \in B \subseteq U$$

#### Example

$X = \mathbb{R}$ ,  $\beta$  = set of open intervals in  $\mathbb{R}$

$X = \mathbb{R}^n$ ,  $\beta$  = set of open balls  $B(x, r) = \{y \in \mathbb{R}^n \mid d(x, y) < r\}$

Check that the second example is actually a basis for a topology

- every  $x \in \mathbb{R}^n$  lies in some open ball.  $x \in B(x, 1)$
- Assume  $x \in B_1 \cap B_2 = B(y_1, r_1) \cap B(y_2, r_2)$   
Set  $r_3 = \min(r_1 - d(x, y_1), r_2 - d(x, y_2))$   
Consider the ball  $B_3 = B(x, r_3)$ . Let  $z \in B_3$ .  
 $d(z, y_1) \leq d(x, y_1) + d(z, x) < d(x, y_1) + r_3 \leq d(x, y_1) + r_1 - d(x, y_1) = r_1$   
So  $z \in B(y_1, r_1) = B_1$  do the same proof for  $B_2$  thus  $z \in B_1 \cap B_2$  thus  $B_3 \subseteq B_1 \cap B_2$

### Proposition 2.2

This is actually a topology on  $X$ .

*Proof.*

- $\emptyset \in T$  vacuously, and  $X \in T$  comes from first basis axiom.
- Suppose  $\{u_j\}_{j \in J}$  of sets in  $T$ .  
Let  $x \in \bigcup_{j \in J} u_j$ . Thus  $x \in u_{j'}$  for some  $j' \in J$ , so  $\exists B_{j'} \in \beta$  s.t.  $x \in B_{j'} \subseteq u_{j'}$  thus  $x \in B_{j'} \subseteq \bigcup_{j \in J} u_j$ . This shows  $\bigcup_{j \in J} u_j \in T$
- Suppose  $x \in u_1 \cap u_2$  for  $u_1, u_2 \in T$ . Then  $x \in B_1, B_2$  for sets  $B_1, B_2 \in \beta$  s.t.  $B_1 \subseteq u_1$

and  $B_2 \subseteq u_2$  thus  $x \in B_1 \cap B_2$ . By the second axiom thus implies  $\exists B_3 \in \beta$ . Thus  $x \in B_3 \subseteq B_1 \cap B_2 \subseteq u_1 \cap u_2 \in T$

**Remark.**

Every topology is a basis for itself.

### Proposition 2.3

Let  $X$  be a set, let  $\beta$  be a basis for topology  $T$ . The sets in  $T$  are precisely the sets of the form

$$\bigcup_{j \in J} B_j$$

for  $\{B_j\}_{j \in J}$  collection of sets in  $\beta$

**Proof.**

- Every set of this form is in  $T$ :

$$x \in \bigcup_{j \in J} B_j, \text{ then } x \in B_{j'} \subset \bigcup_{j \in J} B_j$$

- Every set in  $T$  has the form:

$$\text{Let } U \in T. \text{ Then for every } x \in U, \text{ there is a } B_x \in \beta \text{ s.t. } B_x \subset U \implies U = \bigcup_{x \in U} B_x$$

Decomposition is NOT in general unique.

Basis sets are always open.

## 2.1 Comparing Topologies

### Definition 2.4: Finer

If  $T_1, T_2$  are topologies on the same set  $X$ , we say that  $T_1$ , is finer than  $T_2$  if  $T_1 \supseteq T_2$ . Also,  $T_2$  is coarser than  $T_1$

### Proposition 2.5

Let  $\beta, \beta'$  be bases for topologies  $T, T'$  on  $X$ . Then  $T \subseteq T'$  ( $T'$  is finer than  $T$ ) if and only if every  $x \in X$ , and every  $B \in \beta$  containing  $x$ ,  $\exists$  some  $B' \in \beta'$  s.t.  $x \in B' \subset B$

#### Example

$X = \mathbb{R}^n$ , let  $\beta$  be the set of open boxes in  $\mathbb{R}^n$ : sets of the form  $(a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_n, b_n)$

This is a basis. It generates the same standard topology on  $\mathbb{R}^n$ .

#### Example

Let  $X = \mathbb{R}$ . Consider collections of half-open intervals in  $\mathbb{R}$ :  $[a, b), (a, b]$ .

Claim: this is a basis for a topology.

This is not the same as the std. topology on  $\mathbb{R}$

This is finer than the std. topology.

### 3 Examples of topological spaces

#### *Example 3.1: Order Topology*

Let  $X$  be a set with the order relation  $<$ . For  $a < b$  in  $X$ , define intervals:

$$(a, b) = \{x \in X \mid a < x < b\}$$

$$[a, b) = \{x \in X \mid a \leq x < b\}$$

Similar for  $(a, b], [a, b]$

Let  $\beta$  be the set of all intervals in  $X$  of the forms:

- $(a, b)$  for  $a < b$  in  $X$ .
- $[a_0, b)$  where  $a_0$  is the smallest element of  $X$ .(if it exists).
- $(a, b_0]$  where  $b_0$  is the largest element of  $X$ .(if it exists).

#### **Proposition 3.2**

This is a basis for a topology on  $X$ .

This is called the order topology on  $X$ .

#### *Example*

Standard topology on  $\mathbb{R}$  is order topology, on  $\mathbb{R}$

#### *Example 3.3: Product Topology*

Let  $X, Y$  be topological spaces.

Let  $\beta$  be a collection of sets:

$$\{U \times V \mid U \text{ open in } X, V \text{ open in } Y\}$$

If  $\beta_1$  is a basis for topology on  $X$ , and  $\beta_2$  is a basis for topology on  $Y$ , then

$$\{B \times C \mid B \in \beta_1, C \in \beta_2\}$$

is a basis for the product topology.

**Proposition 3.4**

This is a basis for a topology.

This is called the product topology on  $X \times Y$

**Example**

The product topology is std topology on  $\mathbb{R}$

This definition is valid for products of pairs of sets.

**Lemma 3.5**

Suppose  $X$  is a topological space. Let  $C$  be a collection of open subsets satisfying:

- for each open set  $U \subseteq X$  and each  $x \in U$ ,  $\exists c \in C$  s.t.  $x \in c \subseteq U$

Then,  $C$  is a basis for top. on  $X$ .

**Proof.**

Check  $C$  is a basis.

- Every  $x \in X$  in some  $c \in C$ .  $X$  is open, so there exists  $c$  s.t.  $x \in c \subseteq X$
- If  $c_1, c_2 \in C$ , then for any  $x \in c_1 \cap c_2$ ,  $\exists c_3 \in C$  s.t.  $x \in c_3 \subseteq c_1 \cap c_2$ .  
Follows from the fact that  $c_1, c_2$  are open which implies  $c_1 \cap c_2$  is open so can find  $c_3$

Check:  $C$  generates same topology  $T$  on  $X$ .

First, suppose  $U$  is open in  $T$ . By assumption for all  $x \in U$ , there exists  $c \in C$  s.t.  $x \in c \subseteq U$ , so  $U$  is open in the topology generated by  $C$ .

If  $U$  is open in topology generated by  $C$ . Then  $U$  is a union of sets in  $C$ . But every  $c \in C$  is open in  $T$ , so union  $U$  is also open in  $T$ . ■

**Definition 3.6: Metric**

Let  $X$  be a set. A function  $d : X \times X \rightarrow \mathbb{R}$  is a metric if it satisfies:

- $d(x, y) = 0 \iff x = y$
- $d(x, y) = d(y, x) \quad \forall x, y \in X$
- $d(x, y) + d(y, z) \geq d(x, z) \quad \forall x, y, z \in X$

**Example 3.7: Metric Topology**

A set  $X$  equipped with a metric  $d$  is a metric space.

Let  $\beta$  be the set of all metric balls:

$$B_d(x, r) = \{y \in X \mid d(x, y) < r\} \quad \text{for } x \in X, r > 0$$

**Proposition 3.8**

This is a basis for a topology

**Definition 3.9: Metrizable**

A topological space  $X$  is called metrizable if  $\exists$  metric  $d$  on  $X$  inducing topology on  $X$ .

**Remark.**

There could be more than one metric inducing the same topology.

**Example 3.10: Subspace Topology**

Let  $X$  be a set with topology  $\mathcal{T}$ . Let  $Y \subseteq X$ .

Thus subspace topology on  $Y$  is the collection of sets:

$$\{U \cap Y \mid U \in \mathcal{T}\}$$

**Proposition 3.11:  $\mathcal{T}$** 

is a topology on  $Y$ .

We call  $Y$  a subspace of topology  $X$ .

**Example**

Suppose  $(X, d)$  is a metric space.  $Y \subseteq X$ .

The function  $d$  is also a metric on  $Y$ . We have 2 topologies on  $Y$ :

- Subspace topology as a subspace of  $X$  with metric topology.
- Metric topology on  $Y$

**Proposition 3.12**

These two topologies are the same.

*Proof.*

Let  $\mathcal{T}_{\subseteq}$  is subspace topology, and  $\mathcal{T}_d$  is metric topology on  $Y$ .

$\mathcal{T}_d$  has a basis  $\beta_d = \{B_d(y, r) \cap Y \mid y \in Y, r > 0\}$

$\mathcal{T}_{\subseteq}$  has a basis  $\beta_{\subseteq} = \{B_d(x, r) \cap Y \mid x \in X, r > 0\}$

Subspace Basis is just  $X$ 's basis but intersected with  $Y$ .

Easy to see that  $\beta_d \subseteq \beta_{\subseteq}$  thus  $T_d \subseteq T_{\subseteq}$

Let  $y \in B \in \beta_{\subseteq}$ , so  $y \in B_d(x, r) \cap Y$  for some  $x \in X, r > 0$

Using the triangle inequality, find some  $r' > 0$  s.t.

$$B_d(y, r') \subseteq B_d(x, r)$$

Then

$$B_d(y, r') \cap Y \subseteq B_d(x, r) \cap Y$$

**Example**

Let  $X, Y$  be nonempty topological spaces. Equip  $X \times Y$  with the product topology. For each  $y \in Y$ , the set  $X \times \{y\}$  is a subspace of  $X \times Y$ , and  $X \times \{y\}$  is in bijection with  $X$ . Two topologies on  $X$ :

- Original topology
- Subspace topology on  $X \times \{y\}$  identified with  $X$ .

**Proposition 3.13**

These two topologies are the same.

**Example**

Let  $X$  be a set with an order  $<$ . Let  $Y \subseteq X$ . Then  $<$  gives an order relation of  $Y$ .

Two topologies on  $Y$ :

- Order topology on  $Y$
- Subspace topology

**Proposition 3.14**

These do not have to agree!.

*Proof.*

Let  $X = \mathbb{R} \times \mathbb{R}$ , with dictionary order and  $Y = [0, 1] \times [0, 1]$ . Consider  $\{1/2\} \times [0, 1]$ : This is open in the subspace topology since  $\{1/2\} \times (-\epsilon, 1 + \epsilon)$  is an interval in  $X$ . This is not open on the order topology since it has the endpoints. ■

## 4 Continuous Functions

### Definition 4.1: Continuous Function

Let  $X, Y$  be topological spaces. A function  $f : X \rightarrow Y$  is continuous if, for every open  $U \subseteq Y$ , the set  $f^{-1}(U)$  is open in  $X$ .

### Definition 4.2: Homeomorphism

A bijection  $f : X \rightarrow Y$  is called a homeomorphism if it is continuous and  $f^{-1} : Y \rightarrow X$  is also continuous.

Equivalently: If  $f : X \rightarrow Y$  is a bijection, then it induces a bijection  $P(X) \rightarrow P(Y)$ . If  $X, Y$  are topological spaces, a bijection  $f : X \rightarrow Y$  is a homeomorphism if it identifies topology on  $X$  with the topology on  $Y$ .

#### Example

Let  $X = \{a, b\}$  and  $Y = \{c, d\}$ . Let  $\mathcal{T}_X = \{\emptyset, X, \{a\}\}$ . Let  $\mathcal{T}_Y = \{\emptyset, Y\}$ . Consider  $f : X \rightarrow Y$  s.t. it is a bijection this is clearly continuous since  $f^{-1}(Y) = (X)$ . However for  $f^{-1} : Y \rightarrow X$  this is not continuous.

### Definition 4.3: Embedding

Let  $f : X \rightarrow Y$ , let  $Z \subseteq Y$  be the subspace  $f(X)$ . If  $f : X \rightarrow Z$  is a homeomorphism, then we say that  $f$  is a topological embedding.

This is not the same as a continuous injective map.

#### Example

Let  $X = [0, 2\pi)$ ,  $f : X \rightarrow \mathbb{R}^2$  given by  $f(t) = (\cos(t), \sin(t))$ .

### Proposition 4.4

Let  $f : X \rightarrow Y$  be a map between topological spaces  $X, Y$ .

$$f \text{ is continuous} \iff f^{-1}(B) \text{ is open for every } B \text{ in a basis for } Y.$$

*Proof.*

- $(\implies)$  : immediate by the fact that basis sets are open.
- $(\impliedby)$  : use fact that every open set in  $Y$  is a union of basis sets. ■

### Definition 4.5: Neighborhood

Let  $X$  be a topological space,  $x \in X$ . A neighborhood of  $x$  is any open set  $U \subset X$  containing  $x$ .

### Definition 4.6: Continuous at a Point

A map  $f : X \rightarrow Y$  between topological spaces is continuous at  $x \in X$  if, for every neighborhood  $V$  of  $f(x)$ , there is a neighborhood  $U$  of  $x$  s.t.  $f(U) \subseteq V$ .

### Proposition 4.7

Let  $f : X \rightarrow Y$  be a map between topological spaces then the following are equivalent.

- (i)  $f$  is continuous
- (ii)  $f$  is continuous at every  $x \in X$
- (iii)  $f^{-1}(C)$  is closed for every closed  $C \subseteq Y$

*Proof.*

(i)  $\iff$  (iii) : Let  $U \subseteq Y$  be open.

Thus,  $X \setminus f^{-1}(U) = f^{-1}(Y \setminus U) = f^{-1}(Y \setminus U)$

Thus,  $f^{-1}(U)$  is open  $\iff f^{-1}(Y \setminus U)$  is closed.

(i)  $\implies$  (ii) : Let  $x \in X$ , let  $V$  be a neighborhood of  $f(x)$ . Then  $f^{-1}(V)$  is open, so it's a neighborhood of  $x$  and  $f(f^{-1}(V)) \subseteq V$ .

(ii)  $\implies$  (i) : Let  $V \subseteq Y$ . For each  $x \in f^{-1}(V)$ ,  $f(x) \in V$ , so there is a neighborhood  $U_x$  of  $x$  s.t.  $f(U_x) \subseteq V$ , meaning  $U_x \subseteq f^{-1}(V)$ .

Thus,  $f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$ , so  $f^{-1}(V)$  is open. ■

### Definition 4.8: Continuous on Metric Spaces

Let  $X, Y$  be metric spaces. A function  $f : X \rightarrow Y$  is continuous at  $x \in X$  if  $\forall \epsilon > 0$ ,  $\exists \delta > 0$  s.t. whenever  $d(x, y) < \delta$ , then  $d(f(x), f(y)) < \epsilon$ .

Also the same as, for every ball  $B(f(x), \epsilon)$ ,  $\exists \delta$  s.t.  $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$ .

**Proposition 4.9**

These definition of continuity are equivalent when  $X, Y$  are metric spaces.

**Lemma 4.10**

Let  $f : X \rightarrow Y, g : Y \rightarrow Z$  be continuous between topological spaces. Then  $g \circ f$  is continuous.

*Proof.*

Let  $U \subseteq Z$  be open then  $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$ .  $g^{-1}(U)$  is open, since  $g$  is continuous then  $f^{-1}(g^{-1}(U))$  is open, since  $f$  is continuous. ■

**Proposition 4.11**

Let  $X$  be a topological space,  $A \subseteq X$  a subspace then the inclusion  $f : A \hookrightarrow X$  is continuous.

*Proof.*

Let  $U \subseteq X$  be open, thus  $f^{-1}(U) = U \cap A$  is open in  $A$  by definition of subspace topology. ■

**Proposition 4.12**

Let  $f : X \rightarrow Y$  be a map between topological spaces, suppose that  $f(X) \subseteq Z$  for some subspace  $Z \subseteq Y$ . Then  $f : X \rightarrow Z$  is continuous  $\iff f : X \rightarrow Y$  is continuous.

**Proposition 4.13**

If  $f : X \rightarrow Y$  is continuous, and  $A$  is a subspace of  $X$  then  $f|_A : A \rightarrow Y$  is continuous.

**Definition 4.14: Open Covering**

Let  $X$  be a topological space. An open covering of  $X$  is a collection  $\{U_\alpha\}_{\alpha \in J}$  of open sets s.t.  $X = \bigcup_{\alpha \in J} U_\alpha$ .

**Proposition 4.15**

Let  $f : X \rightarrow Y$ , let  $\{U_\alpha\}_{\alpha \in J}$  be an open covering. Then  $f$  is continuous  $\iff$  each restriction  $f|_{U_\alpha}$  is continuous.

**Proposition 4.16**

Suppose  $X$  is a union of finitely many closed subsets  $C_1, \dots, C_n$ .  
Then a map  $f : X \rightarrow Y$  is continuous  $\iff$  each restriction  $f|_{C_1} : C_1 \rightarrow Y$  is continuous.

Let  $\mathbb{R}$  have std. top.

- Addition
- Multiplication
- Division ( $x/y$ ) for  $y \neq 0$

These are continuous.

**Lemma 4.17**

If  $f_1 : X_1 \rightarrow Y_1$ ,  $f_2 : X_2 \rightarrow Y_2$  are continuous maps of top spaces, then the map  $X_1 \times X_2 \rightarrow Y_1 \times Y_2$  given by  $(u, v) \mapsto (f_1(u), f_2(v))$  is continuous.

## 5 Closures of Sets in Topological Spaces

### Definition 5.1: Closure of a set

Let  $X$  be a topological spaces,  $A \subseteq X$ . The closure of  $A$  in  $X$ , denoted  $\bar{A}$ , is the intersection of all closed subsets of  $X$  containing  $A$ . (The "smallest" closed subset containing  $A$ )

### Definition 5.2: Interior of a set

The interior of  $A$  is the unions of all open subsets contained in  $A$ .

$$\text{Int}(A) = X \setminus \overline{X \setminus A}$$

### Theorem 5.3

Let  $X$  be a top. space, let  $A \subseteq X$ . Then  $x \in \bar{A} \iff$  every neighborhood of  $x$  intersects  $A$ .

*Proof.*

This is the same as  $x \notin \bar{A} \iff$  there is some neighborhood of  $x$  which does not intersect  $A \iff$  there exists a closed subset of  $X$  containing  $A$ , and not  $x$ .

This is equivalent to  $x \notin \bar{A}$  since it is not in some closed set containing  $A$ . ■

### Corollary 5.4

Suppose  $\beta$  is a basis for topology on  $X$ . Then  $x \in \bar{A} \iff$  every  $B \in \beta$  containing  $x$  intersects  $A$ .

*Proof.*

( $\implies$ ) this is proved from the last theorem since basis sets are open.

( $\impliedby$ ) Let  $U$  be a neighborhood of  $x$ . Then, there is some  $B \in \beta$  s.t.  $x \in B \subseteq U$ . Know  $B$  intersects  $A$ , so  $U$  also intersects  $A$ . This means every neighborhood of  $x$  intersects  $A$ , so  $x \in \bar{A}$  ■

**Example:** Let  $X = \mathbb{R}^2$ , let  $A = B(0,1)$  then  $\bar{A} = \bar{B}(0,1) = \{x \in \mathbb{R}^2 \mid d(0,x) \leq 1\}$

First, supposed  $x \notin \bar{B}(0,1)$ . Then  $d(0,x) = r > 1$ . Using the triangle inequality, can see that  $B(x, r-1)$  does not intersect  $B(0,1)$ . Thus,  $x \notin \bar{B}(0,1)$ .

Suppose  $x \in \bar{B}(0,1)$ . Then  $d(0,x) \leq 1$ . Fix  $\epsilon > 0$ . Consider the point  $y = (1-\epsilon)x$ . Thus,  $d(x,y) = d(x, (1-\epsilon)x) = d(x, x - \epsilon x) = d(0, -\epsilon x)$  is given by  $\|\epsilon x\| = \epsilon \|x\| \leq \epsilon$

$y \in B(x, 2\epsilon)$ . Also,  $d(0, y) = d((1 - \epsilon)x, 0) = (1 - \epsilon)\|x\| \leq 1 - \epsilon < 1$ , so  $y \in B(0, 1)$ . Thus, every ball about  $x$  intersects  $B(0, 1)$ , so  $x \in \overline{B(0, 1)}$  ■

**Remark.**

- $\bar{\bar{A}} = \bar{A}$
- $A$  is closed  $\iff A = \bar{A}$

### Definition 5.5: Accumulation Point

Let  $A$  be a subset of a space  $X$ . An accumulation point of  $A$  is a point  $x \in X$  s.t. every neighborhood of  $x$  intersects  $A$  in a point not equal to  $x$ .

**Example**

Every point in  $\mathbb{R}$  is an accumulation point of  $\mathbb{Q}$ .

No points in  $\mathbb{R}$  are accumulation points in  $\mathbb{N}$ .

If  $A = \{1/n \mid n \in \mathbb{N}\}$ , then 0 is the only accumulation point of  $A$ . ■

### Theorem 5.6

Let  $X$  be a topological space, let  $A \subseteq X$ , and  $A'$  be the set of all accumulation points of  $A$ . Then  $\bar{A} = A \cup A'$ .

**Proof.**

Show:  $A \cup A' \subseteq \bar{A}$

Know:  $A \subseteq \bar{A}$ . So let  $x \in A'$

Then every neighborhood of  $x$  intersects  $A$ , so  $x \in \bar{A}$ .

Show:  $\bar{A} \subseteq A \cup A'$ . Let  $x \in \bar{A}$ . If  $x \in A$ , we are done. So, assume  $x \notin A$ . Know every neighborhood of  $x$  intersects  $A$ . Since  $x \notin A$ , this neighborhood must actually intersect  $A \setminus \{x\} = A$  ■

### Definition 5.7: Converges

Let  $x_n$  be a sequence in a space  $X$ . We say that  $x_n$  converges to  $x \in X$  if, for all neighborhoods  $U$  of  $x$ ,  $\exists N \in \mathbb{N}$  s.t.  $x_n \in U$  for all  $n \geq N$ .

**Example**

Let  $X = \{a, b, c\}$ . Let  $\mathcal{T} = \{\emptyset, X, \{a, b\}\}$ .

Consider the sequence  $x_n = a$  converges to  $a$ , since it is in every neighborhood of  $a$ , but

it also converges to  $b$  since  $x_n$  is in every neighborhood of  $b$ . It also converges to  $c$  since  $x_n$  is in every neighborhood of  $c$ . ■

### Example

If  $X$  is a space with indiscrete topology then every sequence converges to every point in  $X$ . ■

## Definition 5.8: Hausdorff Space

A topological space is called Hausdorff if it satisfies the following:

- For every pair of distinct points  $x_1$  and  $x_2$ , there are neighborhoods  $U_1$  of  $x_1$  and  $U_2$  of  $x_2$  s.t.  $U_1 \cap U_2 = \emptyset$ .

## Theorem 5.9

Let  $x_n$  be a sequence in a Hausdorff space. Then  $x_n$  converges to at most one point.

*Proof.*

Suppose  $x_n$  converges to  $x$ . Let  $y$  be any point distinct from  $x$ .

WTS:  $x_n$  does not converge to  $y$ .

Fix neighborhoods  $U$  of  $x$  and  $V$  of  $y$  s.t.  $U \cap V = \emptyset$ . For all but finitely many  $n$ ,  $x_n \in U$ .

This means  $x_n \in V$  for only finitely many  $n$ , so  $x_n$  cannot converge to  $y$ . ■

## Theorem 5.10

Let  $X$  be a Hausdorff space. Then any singleton  $\{x\}$  is a closed subset of  $X$ .

*Proof.*

WTS:  $X \setminus \{x\}$  is open.

For each  $y \in X \setminus \{x\}$ , there is a neighborhood  $U_y$  of  $x$  and  $V_y$  of  $y$  s.t.  $U_y \cap V_y = \emptyset$ . In particular  $V_y \subseteq X \setminus \{x\}$ . Thus:

$$X \setminus \{x\} = \bigcup_{y \in X \setminus \{x\}} U_y$$

is open. ■

In that proof we really only needed: for every pair of distinct points  $x, y \in X$ , there exists a neighborhood  $U$  of  $x$  not containing  $y$ . ( $T_1$  property)

**Proposition 5.11**

Every metrizable space  $X$  is Hausdorff.

*Proof.*

Let  $x, y \in X$ . Set  $r = d(x, y)$ , then by the triangle inequality,  $B(x, r/2) \cap B(y, r/2) = \emptyset$ . ■

**Proposition 5.12**

Let  $X$  be a top. space, let  $A \subseteq X$ . Suppose  $x_n$  is a sequence in  $A$ , and  $x_n \rightarrow x$  in  $X$ . Then  $x \in \bar{A}$

*Proof.*

Let  $U$  be a neighborhood of  $x$ . Then  $x_n \in U$  for all but finitely many  $n$ , so  $U \cap A \neq \emptyset \implies x \in \bar{A}$  ■

**Proposition 5.13**

Assume  $X$  is metrizable. Then, every  $x \in \bar{A}$  is a limit of a sequence  $x_n$  for  $x_n \in A$

*Proof.*

Let  $x \in \bar{A}$ . For every  $n \in \mathbb{N}$ , the ball  $B(x, 1/n)$  intersects  $A$ . So, let  $x_n \in A \cap B(x, 1/n)$ . If  $U$  is any neighborhood of  $x$ , then  $U$  contains  $B(x, \epsilon)$  for some  $\epsilon > 0$ . Then  $x_n \in B(x, \epsilon)$  whenever  $n > 1/\epsilon$ . Thus,  $x_n \rightarrow x$ . ■

**Theorem 5.14**

Let  $f : X \rightarrow Y$  be a map of topological spaces. Then  
 $f$  is continuous  $\implies$  for any sequence  $x_n$  in  $X$  converging to a point  $x \in X$  we have  $f(x_n) \rightarrow f(x)$   
 If  $X$  is metrizable  $\iff$  holds.

## 6 Quotient Topology

Let  $X$  be a top. space, let  $\sim$  be an equivalence relation on  $X$ . There is a surjective map  $\rho : X \rightarrow X/\sim$   $x \mapsto [x]$

$X/\sim$  is the set of equivalence classes with respect to  $\sim$ .

### Definition 6.1: Quotient Topology

The quotient topology on  $X/\sim$  is the collection of sets  $U \subseteq X/\sim$  s.t.  $\rho^{-1}(U)$  is open in  $X$ .

This is the finest topology on  $X/\sim$  s.t.  $\rho : X \rightarrow X/\sim$  is continuous.

### Definition 6.2: Quotient Map

In general, if  $X, Y$  are top. spaces, a surjective map  $\rho : X \rightarrow Y$  is a quotient map if it satisfies:

$$U \subseteq Y \text{ open} \iff \rho^{-1}(U) \text{ open}$$

When this occurs,  $Y$  can be identified homeomorphically to a quotient space for  $X$ , under the relation  $x_1 \sim x_2 \iff \rho(x_1) = \rho(x_2)$

If  $X, Y$  are top. spaces, can form disjoint union  $X \sqcup Y$ . Topology on  $X \sqcup Y$  is the collection  $U \sqcup V$  for  $U \subseteq X, V \subseteq Y$  open.

Suppose  $X_1, X_2$  spaces,  $Y_1 \subseteq X_1, Y_2 \subseteq X_2$  subspaces,  $\phi : Y_1 \rightarrow Y_2$  is a homeomorphism.

Can form space  $X_1 \sqcup_{\phi} X_2 = X_1 \sqcup X_2 / \sim_{\phi}$

$\sim_{\phi}$  is relation given by:  $x \sim_{\phi} x$  and  $y \sim_{\phi} \phi(y)$  and  $\phi(y) \sim_{\phi} y$  for every  $y \in Y_1$

### Theorem 6.3

Suppose  $f : X \rightarrow Y$  is map, and  $\sim$  is an equivalence relation. There is a map  $X/\sim \rightarrow Y$  is  $f(x) = f(y)$  whenever  $x \sim y$

If  $X/\sim$  has the quotient topology, then  $f$  is continuous  $\iff f' : X/\sim \rightarrow Y$  is continuous.

## 7 Product Spaces (revisited)

Let  $\{X_\alpha\}_{\alpha \in J}$  be a collection of sets. Can form Cartesian product  $\prod_{\alpha \in J} X_\alpha$ .

If  $X_\alpha = X$ , denoted  $X^J \iff \text{map: } J \rightarrow X$

If each  $X_\alpha$  top spaces, first guess for topology on  $\prod_{\alpha \in J} X_\alpha$

### Definition 7.1: Box topology

Topology generated by the basis  $\prod_{\alpha \in J} U_\alpha$  s.t. each  $U_\alpha$  open in  $X_\alpha$ .

We have natural maps associated to product:

$$\pi_\alpha : \prod_{\alpha \in J} X_\alpha \rightarrow X_\alpha$$

for each  $\alpha \in J$  called projection maps.

### Definition 7.2: Product Topology

If  $\{X_\alpha\}_{\alpha \in J}$  family of top. spaces, product topology on  $\prod_{\alpha \in J} X_\alpha$  is the coarsest topology s.t. all projections are continuous.

This is in general **NOT** the same as box topology.

A subbasis for product topology is collection of all sets of the form  $\pi_\alpha^{-1}(U_\alpha)$  for  $U_\alpha$  open in  $X_\alpha$ .

Each  $\pi_\alpha^{-1}(U_\alpha)$  is a set of the form  $\prod_{\beta \in J} U_\beta$ , where all  $U_\beta$  for  $\alpha \neq \beta$  are  $X_\beta$ .

Intersection of finitely many elements in subbasis looks like  $\prod_{\alpha \in J} U_\alpha$ , where  $U_\alpha = X_\alpha$  for all but finitely many  $\alpha$ .

Sets of this form are a basis for product topology. No way to write a set of the form  $\prod_{\alpha \in J} U_\alpha$

where  $U_\alpha \subsetneq X_\alpha$  for all  $\alpha$  as a union of these basis elements. This implies box topology is finer than product topology.

Let  $\{X_\alpha\}_{\alpha \in J}$  be a family of spaces, for each  $\alpha \in J$ , let  $f_\alpha : Y \rightarrow X_\alpha$  be a map for some spaces.

Defines a map  $f : Y \rightarrow \prod_{\alpha \in J} X_\alpha$  by  $f(y) = (f_\alpha(y))_{\alpha \in J}$

### Proposition 7.3

$f$  is continuous with respect to product topology on  $\prod_{\alpha \in J} X_\alpha \iff$  every  $f_\alpha$  is continuous.

*Proof.*

$\implies$  : By definition of product topology  $\pi_\alpha$  is continuous, and  $f_\alpha = \pi_\alpha \circ f$ , so this is continuous.

$\impliedby$  : Suffices to prove that  $f^{-1}(U)$  is open for every  $U$  of the form  $\pi_\alpha^{-1}(U_\alpha)$  for some  $\alpha \in J$ ,  $U_\alpha \subseteq X_\alpha$  open.

$$f^{-1}(U) = f^{-1}(\pi_\alpha^{-1}(U_\alpha)) = f_\alpha^{-1}(U_\alpha)$$

open since  $f_\alpha$  is continuous. ■

## 7.1 Topologies on $\mathbb{R}^J$ , where $J$ is infinite

### Proposition 7.4

Let  $(X, d)$  be a metric spaces. The function  $\bar{d} : X \times X \rightarrow \mathbb{R}_{\geq 0}$  by  $\bar{d}(x, y) = \min(d(x, y), 1)$  is a metric on  $X$ , induces same topology as  $d$ .

### Definition 7.5: Diameter

The Diameter of a metric space  $(X, d)$  is  $\sup\{d(x, y)\}$  for all  $x, y \in X$

### Definition 7.6: Uniform Metric

Let  $J$  be an index set. Define metric  $\bar{\rho}$  on  $\mathbb{R}^J$  by

$$\bar{\rho}(x, y) = \sup_{\alpha \in J} \bar{d}(x_\alpha, y_\alpha)$$

Induces uniform topology.

When  $J$  is finite this is the same topology as product topology on  $\mathbb{R}^n$

Otherwise: Uniform Topology lies strictly between the product and box topologies on  $\mathbb{R}^J$ .

### 7.1.1 Product Topology on $\mathbb{R}^\omega$

#### Theorem 7.7

The function  $D : \mathbb{R}^\omega \times \mathbb{R}^\omega \rightarrow \mathbb{R}_{\geq 0}$  given by

$$D(x, y) = \sup_{i \in \mathbb{N}} \frac{\bar{d}(x_i, y_i)}{i}$$

is a metric on  $\mathbb{R}^\omega$  which induces product topology.

Fix ball  $B_D(x, r)$ . Choose  $N \in \mathbb{N}$  s.t.  $1/N < r$ . Then  $B_{\bar{d}}(x_1, r) \times B_{\bar{d}}(x_2, r) \times \dots, B_{\bar{d}}(x_N, r) \times \mathbb{R} \times \mathbb{R} \times \dots$  is open in product topology. It's contained in  $B_D(x, r)$ , since each  $y$  in the product satisfies:

$$\bar{d}(x_i, y_i)/i \leq \bar{d}(x_i, y_i) < r \text{ if } i \leq N$$

$$\bar{d}(x_i, y_i) \leq 1/N < r \text{ if } i > N$$